

Living Without Color: A Probe Study of Grayscale Smartphone Use

MUO-C3030 - Empiirinen tutkimus

Aku Paloheimo 909282

Table of contents

	Living Without Color: A Probe Study of Grayscale Smartphone Use	1
	MUO-C3030 - Empiirinen tutkimus	1
1	Introduction.....	3
2	Previous work.....	3
2.1	Smartphone Use and Digital Wellbeing	3
2.2	Interface Design and Visual Attention.....	4
2.3	Grayscale as a Digital Wellbeing Intervention	4
2.4	Probes for Understanding Everyday Technology Use.....	4
3	Methods.....	5
3.1	Participants.....	5
3.2	Procedure	6
3.3	Data analysis	6
4	Results.....	7
4.1	Daily use of grayscale	7
4.2	Emotional Responses	10
4.3	Adaptation and Disengagement Over Time.....	10
4.4	Post-Study Reflections and Attitudes Toward Future Use	11
5	Discussion.....	12
5.1	Limitations	12
	References.....	14

1 Introduction

Smartphones have become an integral part of everyday life, supporting communication, entertainment, information access, and productivity (International Organization for Standardization, 2019). However, their design often encourages frequent and habitual use (Ting et.al., 2020). Many applications rely on visually engaging interfaces, bright colors, and notifications to attract and maintain user attention (Dekker & Baumgartner., 2024; Holte, et.al, 2023; Kurt & Osueke, 2014). While these design features can improve usability and engagement, they may also contribute to habitual or excessive smartphone use (Ting et al., 2020; James et.al., 2023).

As concerns about digital wellbeing have increased, various strategies have been proposed to help users manage their phone usage (Dekker & Baumgartner., 2024). One simple intervention that has gained attention is the use of grayscale mode, which removes color from the smartphone display. The idea behind this approach is that reducing visual stimulation may make applications, particularly social media, less visually appealing, potentially reducing impulsive engagement (Dekker & Baumgartner., 2024).

Despite the popularity of this idea in digital wellbeing discussions, there is limited research examining how grayscale mode affects real-world smartphone use and user experiences over time (Dekker & Baumgartner., 2024).

Therefore, this study explores how using a grayscale smartphone display influences everyday phone use through a probe study approach. Participants enabled grayscale mode on their smartphones for a period of 6 days and documented their experiences through daily reflections and contextual probes. By collecting data in participants' natural environments, the study aims to capture situated experiences of smartphone interaction, including moments of habitual use, changes in engagement with applications, and users' reflections on the role of color in digital interfaces.

The goal of this study is to answer the following research question: How does using a grayscale smartphone display influence the everyday smartphone usage of young adults? Through this research question the paper will contribute a better understanding of how simple interface modification, removing color from the smartphone display, affects users' behaviors and perceptions of their phones in everyday life.

2 Previous work

2.1 Smartphone Use and Digital Wellbeing

Smartphones are designed to support frequent interaction and sustained engagement. Many applications rely on persuasive design strategies such as notifications, infinite scrolling, and visually engaging interfaces to encourage continued use (Rixen et al., 2023). While these features can enhance usability and entertainment value, they may also contribute to habitual or compulsive patterns of phone use (Monge Roffarello & De Russis, 2019; Rixen et al., 2023).

Concerns about excessive smartphone use have led to increased interest in digital wellbeing tools that help users manage their interaction with technology (Monge Roffarello & De Russis, 2019). These tools include screen time tracking, app usage limits, and notification management systems. Research has shown that increasing users' awareness of their own technological use can sometimes encourage more

mindful behavior, although it still remains unclear whether this impacts user behavior long term (Monge Roffarello & De Russis, 2019).

2.2 Interface Design and Visual Attention

Visual design plays a central role in how users interact with digital interfaces. Color is known to influence attention, emotional response, and perceived attractiveness of interface elements (Kurt & Osueke, 2014). Bright and saturated colors are often used in interface design to highlight important elements such as buttons, notifications, and icons (Kurt & Osueke, 2014).

In mobile applications, color can therefore contribute to making interfaces more engaging and visually rewarding (Kurt & Osueke, 2014; Ting et.al., 2020). Social media applications, for example, often rely on colorful images, videos, and interface elements that capture attention and encourage prolonged interaction (Ting et.al., 2020).

Removing color from an interface may change how users perceive and interact with applications. A grayscale display reduces visual contrast between many interface elements and may make certain content, such as images or videos, less engaging (Holte, et.al, 2023).

2.3 Grayscale as a Digital Wellbeing Intervention

Grayscale smartphone use has been suggested as a strategy for reducing smartphone distraction and encouraging more intentional use (Dekker & Baumgartner., 2024). The approach has gained attention in discussions about digital minimalism and attention management, where users experiment with grayscale displays to make their phones feel less stimulating (Holte, et.al, 2023).

However, relatively little empirical research has examined how grayscale mode influences everyday smartphone interaction over time. There is limited understanding of how users experience grayscale phones in real-world contexts and how it affects habitual behaviors such as automatic app checking (Dekker & Baumgartner., 2024; Holte, et.al, 2023).

Because these experiences are highly contextual and subjective, qualitative approaches are well suited to exploring them. Probe studies allow researchers to collect reflections and observations from participants in their everyday environments, capturing interactions that may not be easily reproduced in controlled laboratory settings (Mattelmäki, 2006).

2.4 Probes for Understanding Everyday Technology Use

Probe-based research methods are commonly used in human–computer interaction (HCI) to explore people’s experiences with technology in everyday contexts (Mattelmäki, 2006). Cultural probes and diary studies allow participants to document their interactions, reflections, and experiences over time (Mattelmäki, 2006; Gaver et al., 1999).

These methods are particularly useful for studying habitual behaviors and subjective experiences, as they encourage participants to capture moments as they occur in their daily lives. Rather than focusing solely on quantitative metrics, probe-studies provide rich qualitative insights into how people interact with technologies in real-world situations (Mattelmäki, 2006).

In this study, a probe-based approach is used to investigate how grayscale mode affects smartphone interaction across different contexts and situations. By combining daily reflections, contextual screenshot probes, and interviews, the study aims to capture a detailed picture of how participants experience grayscale smartphone use over time.

3 Methods

This study used a probe research design to explore how using a grayscale smartphone display influences everyday phone usage habits and user experiences. Probe studies are designed to collect rich, contextual data about participants' daily behaviors and reflections over time (Mattelmäki, 2006). In this study, grayscale mode was used as a behavioral intervention, allowing participants to experience their smartphone interfaces without color for an extended period.

Participants were asked to enable grayscale mode on their smartphones and document their experiences through daily diary entries and contextual screenshot probes. The study also included pre-study and post-study interviews to capture participants' expectations and reflections before and after the probe period.

3.1 Participants

A total of five participants (see Table 1) were recruited from the university community and personal networks through convenience sampling. Eligibility criteria required participants to be daily smartphone users, willing to enable grayscale mode for the duration of the study, and able to complete short daily diary entries. Both Android and iOS users were eligible to participate.

Participants were in their mid to late twenties, meaning they had used smartphones for most of their lives. Three participants used iOS devices and two used Android. Prior to the study, participants reported an average daily smartphone use of 4 hours, with usage concentrated primarily in the evenings. Commonly reported phone activities included social media browsing, messaging, watching videos, and listening to music or podcasts. All five participants had previously attempted to reduce or manage their phone use through strategies such as app time limits or self-imposed restrictions. Participants were already aware that grayscale mode was available on their device, and two had used it prior to the study.

All participants were informed about the study procedure and provided informed consent before participating. Data collected throughout the study was anonymized.

Participant	Device	Avg. Daily use	Primary activities	Used grayscale before
P1	iOS	2-4 hrs.	Messaging, video	Yes
P2	iOS	2-4 hrs.	Social media, messaging, video, music	No
P3	Android	4-6 hrs.	Messaging, video, music	Yes
P4	Android	2-4 hrs.	Social media, messaging, video, music	No
P5	iOS	4-6 hrs.	Social media, messaging, video, music	No

Table 1. List of participants' pre study survey results

3.2 Procedure

The study consisted of three phases: a pre-study survey, a six-day probe period, and a post-study survey.

At the beginning of the study, participants completed a pre-study survey designed to gather background information about their typical smartphone use. This included questions about average daily usage, experiences with distraction or automatic phone checking, and any prior awareness or use of grayscale display settings. Following the survey, participants were guided through the process of enabling grayscale mode on their devices and instructed to keep it enabled throughout the study whenever possible.

During the probe period, participants used their smartphones in grayscale mode for six consecutive days and completed a short daily diary entry. Each entry asked participants to reflect on their perceived amount of phone use relative to their normal baseline, whether grayscale had influenced their behavior that day, how tempted they had felt to disable it, and a brief description of the most noticeable experience of the day. On Days 2 and 4, participants were asked additional questions aimed at capturing a deeper understanding of their experiences. Specifically the questions probed, moments when grayscale felt particularly noticeable or unusual, and instances of automatic phone checking.

In addition to the diary entries, participants were asked to submit one or two screenshots per day of moments when grayscale appeared to influence their phone use. Each screenshot was accompanied by a short-written reflection describing the context of the moment, such as what they had been doing immediately before opening the app. These screenshot probes were intended to ground participants' reflections in concrete, real-time interactions and to capture experiences that might otherwise be difficult to recall retrospectively.

Upon completing the probe period, participants filled out a post-study survey covering their overall experience of using grayscale mode and any perceived changes in their smartphone use over the course of the study. The survey asked about changes in usage frequency and duration, shifts in app engagement, awareness of automatic phone use, and emotional responses such as enjoyment or frustration. A further set of questions explored attitudes toward longer-term use, asking participants to consider in which situations they might use grayscale in the future, how they would feel if their phone were always in grayscale, and whether they intended to continue using it after the study. The survey concluded with an open-ended reflection question inviting participants to describe what the experience had taught them about their own smartphone use.

3.3 Data analysis

The study generated several types of qualitative data, including pre-study survey responses, daily diary entries, screenshot probes and their associated written reflections, and post-study survey responses. All data was collected digitally via an online form.

The data was analyzed using qualitative thematic analysis. The process began with a thorough reading of all collected material to develop an overall understanding of participants' experiences. Recurring patterns and themes were then identified across diary entries, screenshot reflections, and survey responses, with particular attention to how participants described their experience of using a grayscale phone in everyday

contexts. Similar observations were grouped into thematic categories, such as reduced engagement with visually rich applications, increased awareness of habitual phone checking, and frustration with the practical limitations of grayscale, and experiences were compared across participants to identify both shared patterns and individual differences. Additionally, the quantitative data, such as perceived phone use on a fixed scale was analyzed with the help of graphs, to recognize trends more easily.

4 Results

The results are organized around four themes that emerged across the pre-study survey, daily diary entries, and post-study reflections: daily phone use and grayscale's behavioral effects, emotional and experiential responses, adaptation over time, and attitudes toward future use. Findings are presented at the individual level, as the small sample size of this probe study was intended to generate qualitative insight rather than generalizable patterns.

4.1 Daily use of grayscale

As seen in Figure 1 daily self-reported phone use ratings (on a scale of 1–5, where lower scores indicate less use than usual) varied considerably across participants and across the six-day study period. P2 showed the most consistent reduction, reporting ratings of 2 across the first four days before returning to 3 in the final two days. P4 and P1 showed a similar early reduction to a low of 1, but then a quick spike back to a self-assessed 4, showing that the effects of grayscale may have worn off over time. P5's ratings were more variable, oscillating between 2 and 4 throughout the study without a clear directional trend. P3 was also an outlier, reporting a consistent rating of 3 on every single day of the study, suggesting that grayscale had no measurable effect on their overall phone use at the behavioral level.

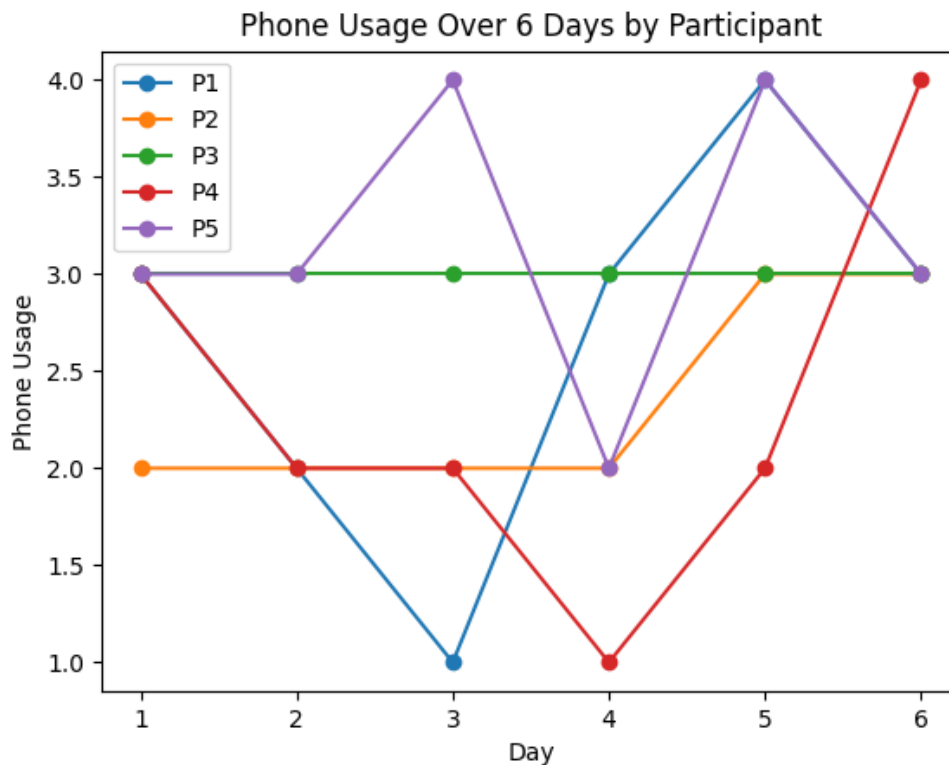


Figure 1. Showing participants’ self-evaluated phone usage over the 6 day probe period (on a scale of 1–5, where lower scores indicate less use than usual)

As seen in Figure 2, whether grayscale was perceived as affecting phone use on a given day also varied. However, there can be some trends found from the data. P2, P3 and P4 reported that grayscale affected their use on four out of the six days. While the timings between the felt affect was different, this suggest some correlation between their experiences. Additionally, participants showed that there may have been some effect during most of the study, as the “No” answer had the least amount of data points of the three options. The participants reported a significant amount of uncertainty however. P5 was the most uncertain, reporting "not sure" on four of the six days and confirming an effect only once and P3 the least, not selecting “not sure” once. Notably, all participants reported feeling tempted to turn grayscale off on at least some days, with P1 and P3 reporting this temptation on every single day of the study.

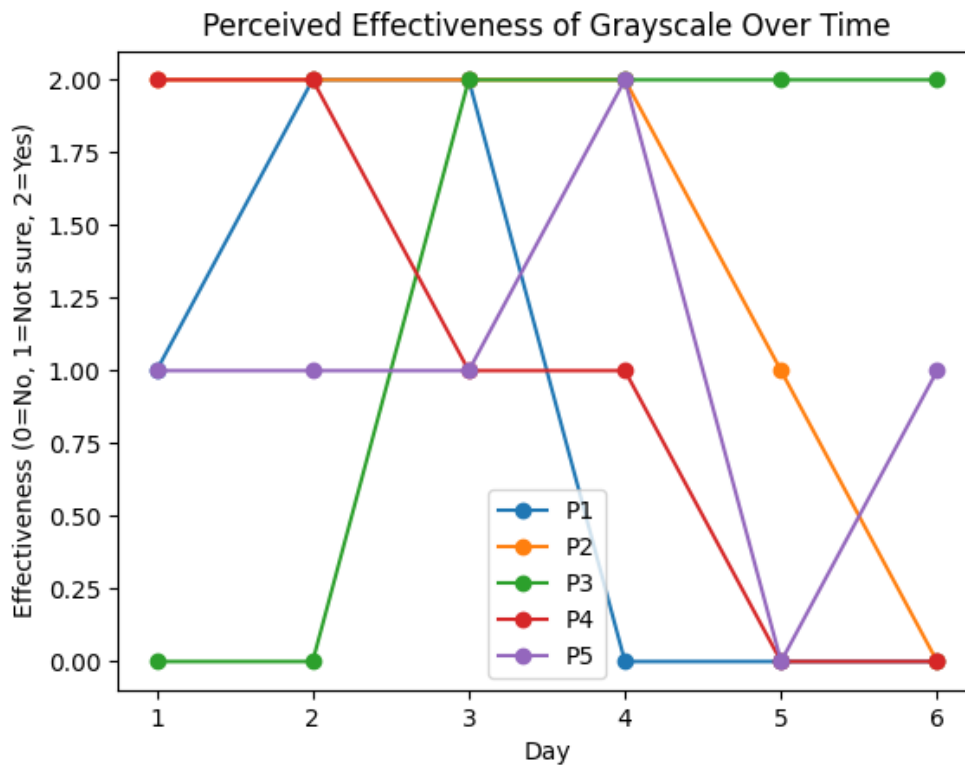


Figure 2. Showing participants’ perceived effectiveness of grayscale usage for each day of the probe period

Several participants described moments where grayscale disrupted their habit of automatic phone checking. P4 noted on the second that they "grabbed my phone while watching something, like I usually do, but then noticed I wasn't even interested in it" a pattern also observed by P2, who described opening the phone without thinking but finding that seeing their phone in grayscale made them feel less interested in what was on the phone. P5 similarly noted that they had "opened social media without thinking, but it felt kinda pointless so I didn't stay long." These moments of disengagement were most common in the early days of the study and were not noted by the participants in the latter phases of the experiment.

The effect of grayscale was most pronounced for visually engaging, content-driven applications. Video platforms and social media were often cited by the participants as feeling less engaging, while applications with more text based interfaces, such as messaging apps, were described less so. P2 noted early on that on Instagram the unseen “stories” (Instagram’s functionality of showing images only for a 24-hour period), normally marked by a colored ring, were less visible, reducing the impulse to check them. P5 noted a shift in application usage over time, observing on the fifth day that they had begun spending more time reading news rather than scrolling Instagram, suggesting that grayscale may have redirected engagement toward less visually dependent content rather than eliminating phone use altogether, but showing the users fixation towards the “endless scrolling” -mechanic, as described by Rixen et al. (2023).

Grayscale also interfered with everyday visual tasks with multiple participants. P1 and P3 noted on separate days the difficulty of viewing photographs properly without color. P3 reported difficulty finding a music album due to the lack of color cues in the

interface and was forced to disable grayscale. P2 found WhatsApp harder to read and expressed frustration that they could not tell whether their messages had been read. P2 also indicated difficulties in adapting to the use of greyscale as they reported they had difficulties finding the apps that they regularly used.

4.2 Emotional Responses

Participants described the experience of using their phones under grayscale in broadly similar terms, though with differences in how they emotionally related to these changes. The recurring theme was a reduced sense of reward or pleasure associated with phone use. P2 described the phone as feeling "not nearly as satisfying or enjoyable" under grayscale, noting that it sometimes felt pointless to use the phone at all when it no longer produced the same sense of pleasure. P4 expressed this most directly in the early days of the study, stating that grayscale had "really affected" their use because they simply did not want to look at things as much, framing this as a helpful intervention for what they described as their "phone addiction."

P5 offered a somewhat different perspective, reporting that putting their phone in grayscale prompted them to use their phone with more purpose; on the second day they noted spending time on their phone "more purposefully instead of just scrolling aimlessly," and by the fourth day described making a conscious effort to keep the phone in grayscale as a way of staying more alert. On the contrary P1 expressed more frustration with the lost functionality of their phone. While they also stated that they had less desire to consume media throughout the experiment, they also described frustration with the practical issues of the intervention, particularly the repeated need to disable and enable grayscale.

P3's emotional response was notably different from the others. Rather than finding the intervention either liberating or frustrating, P3 appeared to adapt relatively quickly. On the fourth day P3 reported that they had "started to adapt to the grayscale and just watched videos in grayscale," indicating that the reduced reward response had faded without producing a big change in behavior.

4.3 Adaptation and Disengagement Over Time

A consistent pattern across participants was a gradual erosion of grayscale's effect over the course of the study. This was driven by the practical ease of disabling the feature. P1 articulated this most clearly, noting that they had become "even more lazy with keeping the grayscale on," estimating that they turned it off approximately half the time they used their phone, and often forgot to re-enable it afterward. By the last day, P1 had effectively stopped using grayscale for most of the day. P1 also reflected that the low friction of toggling grayscale off had fundamentally undermined its effect: "since it's quite easy to turn it off I tend to do it whenever I feel like doing something that is slightly enhanced by not having the grayscale on."

P4 described a similar trajectory, stating on the last day that "grayscale lost its effect a bit" after they discovered a quick way to toggle it off, and found themselves turning it off frequently as a result. P3's disengagement was more gradual, they reported on the last day that they were "constantly" wanting to turn grayscale off, and that it was "not on much" by the end of the study. P2 and P5 showed more varied patterns: P2 stopped feeling tempted to disable grayscale from the third day onward, while P5 alternated between days of high and low temptation without a clear trend.

Across all participants, situations involving visual content were the most common triggers for turning off grayscale. Once disabled, participants frequently forgot to enable it again, contributing to declining adherence over the second half of the study.

Also, it should be noted that most phones have a quick settings menu that can be utilized to ease turning the grayscale on or off, when compared to going through the settings as seen in Figure 3. The participants were not instructed to use these faster methods to see how they would adapt to the use of grayscale

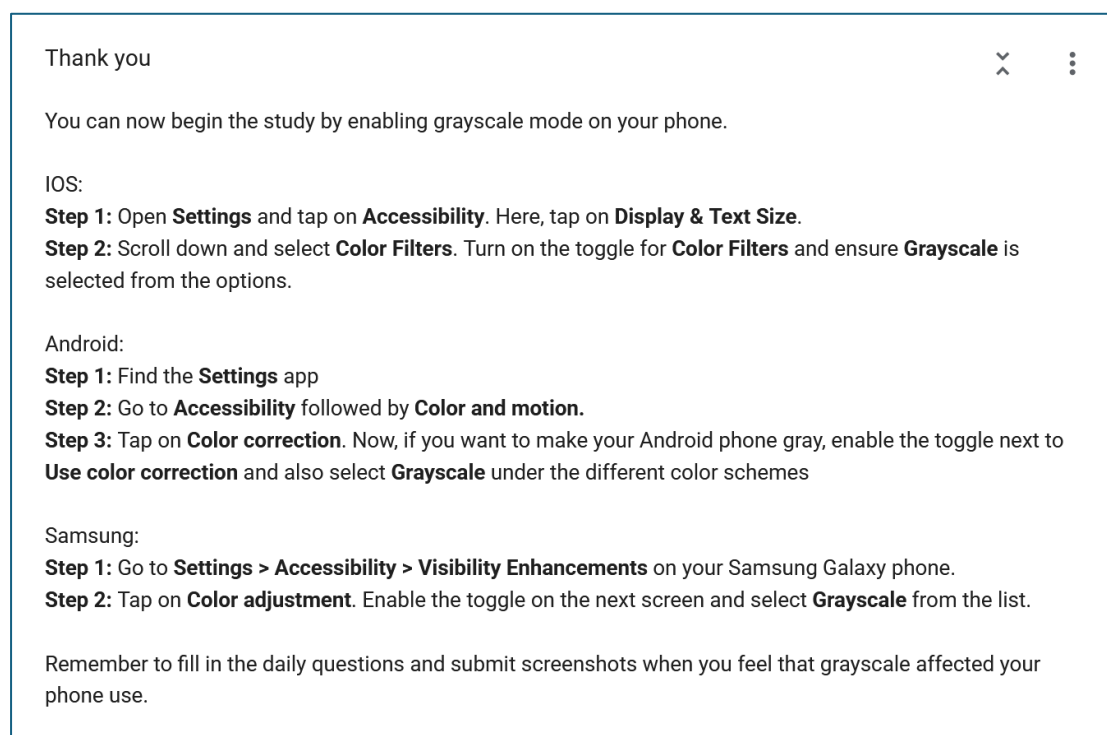


Figure 3. Showing the instructions given to the participants on how to turn on the grayscale filter on most popular phones in use.

4.4 Post-Study Reflections and Attitudes Toward Future Use

Participants' overall assessments of grayscale at the end of the study reflected the varied experiences reported throughout. P2 and P4 reported the most favorable outcomes, with P2 describing a clear reduction in both the frequency and duration of phone use, and P4 similarly reporting using the phone less often. P1 and P3 reported no overall change in usage, while P5's experience was a bit more inconclusive, but additionally did not see a huge change in their behavior.

Attitudes toward future use were mixed. P1 indicated they would not continue with grayscale, finding that the practical inconveniences outweighed the benefits and that their learned habits during the experiment made the effect less impactful. P1 did, however, suggest that a version of the feature with an automatic re-engagement of grayscale, might make it more suitable for their habits. P4 reached a similar conclusion, noting that grayscale was effective early on but that its impact diminished once a way to quickly toggle it off was found. P2 expressed willingness to use grayscale again during periods of perceived overuse, ideally alongside app time limits. P5 indicated that the experience had shifted their phone use toward more deliberate, reading-focused activity, and found value in this even if the effect was not entirely due to grayscale itself.

All participants had a shared takeaway, that grayscale functions more effectively as a tool for making one more aware of their phone use but not the primary way to reduce screen time. P1 reflected that the study had "highlighted that some of my phone use habits are very strong and I'm not that willing to change those habits." P4 noted that while grayscale made the phone "look worse," it did not change how much they used it. On the contrary, P5 felt that the experience had helped them become more intentional by using their phone "for more specific things rather than just mindless scrolling." These differing outcomes suggest that the value of grayscale depends heavily on individual motivation and willingness to keep it on.

5 Discussion

This paper set out to answer the following question: How does using a grayscale smartphone display influence the everyday smartphone usage of young adults?

The findings suggest that grayscale mode can meaningfully influence smartphone use, primarily by reducing the sort of reward a person receives associated with visually stimulating media or applications. Participants frequently reported greater awareness of their phone use and described moments where grayscale appeared to disrupt otherwise automatic behavior. Taken together, these observations support the claim that grayscale can alter the experience of interacting with a device.

A recurring theme across participants was the role of color in keeping engagement. Applications that relied heavily on images and video were consistently described as less attractive without color. Showing that the use of color in user interfaces contributes not only to usability but to the enjoyment of interaction. This has implications for how digital engagement is understood: if color functions as a form of low-level reinforcement or reward, its removal or limitation in interface design may help users limit their phone use and help retain their attention for other tasks.

That said, the effects of grayscale were not uniform. Some participants reported clear changes in their behavior, while others experienced little impact at all. This variation likely comes from differences in individual usage habits, and the specific applications participants used. Grayscale therefore cannot be treated as an intervention that will work for everyone. As, its utility appears to be influenced a lot by the user and the context which it gets used in. Additionally, several participants reported that the initial effect of grayscale dwindled as the participants adjusted to it, which raises questions about its long-term effectiveness.

This paper contributes to ongoing discussions about digital wellbeing by demonstrating that small, accessible changes to interface design can prompt meaningful shifts in awareness and behavior. Grayscale mode is easy to enable, requires no external system, and appears capable of creating at least a temporary disruption to habitual patterns of use. At the same time, the findings make clear that its effectiveness is context-dependent, and that it is best understood as one component of a broader approach to managing digital engagement rather than a direct solution.

5.1 Limitations

Several limitations should be noted. The sample size was relatively small, and the study duration limited to approximately a week. Both of which constrains the generalizability of the findings. Also, data relied primarily on self-report, making

mistakes and bias possible in the study. It is also possible that participation in the study itself heightened awareness of phone use in ways that influenced behavior independently of the grayscale intervention. Additionally, it should be noted that participants may deliberately withhold some of their findings or fabricate their results to evade judgement, as they are aware that their inputs are going to be read and used in research. Even though it is stated that their results are anonymous and that any judgement or preconceptions are not made from their provided data. Also the participants were extremely willing to use grayscale and try to change their behavior and had previously noticed their phone use as an issue. This made the participants hyper aware of their phone use and might mean that not so willing participants might experience grayscale as a more radical intervention.

Additionally, the participants were all young adults that had extensive experience of phones and user interfaces. This limited the findings to this specific group of people and most likely some of the struggles that the participants had with the user interfaces were not as pronounced as with other groups of people. Future studies could focus more on elderly people or people that are not as familiar with the use of a phone and might struggle more when visual cues, like color are removed from the interfaces that they have to use.

Future research would benefit from longer study periods, the inclusion of objective screen time data alongside self-report measures, and a broader range of participant profiles. It would also be valuable to examine how grayscale interacts with other digital wellbeing interventions, and to identify more precisely the contexts and user characteristics for which it is most effective.

References

- Dekker, C. A., & Baumgartner, S. E. (2024). Is life brighter when your phone is not? The efficacy of a grayscale smartphone intervention addressing digital well-being. *Mobile Media & Communication*, 12(3), 688-708.
- Kiss, L., & Linnell, K. J. (2021). The effect of preferred background music on task-focus in sustained attention. *Psychological research*, 85(6), 2313-2325.
- Gaver, B., Dunne, T., & Pacenti, E. (1999). Design: cultural probes. *interactions*, 6(1), 21-29.
- Holte, A. J., Giesen, D. T., & Ferraro, F. R. (2023). Color me calm: Grayscale phone setting reduces anxiety and problematic smartphone use. *Current Psychology*, 42(8), 6778-6790.
- International Organization for Standardization. (2019). Ergonomics of human-system interaction -- Part 210: Human-centred design for interactive systems (ISO Standard No. 9241-210). Retrieved from <https://online.sfs.fi/fi/index/tuotteet/SFS/CENISO/ID2/9/807069.html.stx#>
- James, R. J., Dixon, G., Dragomir, M. G., Thirlwell, E., & Hitcham, L. (2023). Understanding the construction of 'behavior' in smartphone addiction: A scoping review. *Addictive behaviors*, 137, 107503. Retrieved from <https://doi.org/10.1016/j.addbeh.2022.107503>
- Kurt, S., & Osueke, K. K. (2014). The Effects of Color on the Moods of College Students. *Sage Open*, 4(1).
- Mattelmäki, T. (2006). *Design probes*. Aalto University. ISBN : 951-558-211-3
- Monge Roffarello, A., & De Russis, L. (2019, May). The race towards digital wellbeing: Issues and opportunities. In *Proceedings of the 2019 CHI conference on human factors in computing systems* (pp. 1-14). Retrieved from <https://doi.org/10.1145/3290605.3300616>
- Rixen, J. O., Meinhardt, L. M., Glöckler, M., Ziegenbein, M. L., Schlothauer, A., Colley, M., ... & Gugenheimer, J. (2023). The loop and reasons to break it: Investigating infinite scrolling behaviour in social media applications and reasons to stop. *Proceedings of the ACM on Human-Computer Interaction*, 7(MHCI), 1-22. Retrieved from <https://doi.org/10.1145/3604275>
- Ting, C. H., & Chen, Y. Y. (2020). Smartphone addiction. In *Adolescent addiction* (pp. 215-240). Academic Press. Retrieved from <https://doi.org/10.1016/B978-0-12-818626-8.00008-6>